

Bilateral Deficiency: Residual SAT Optimisation and Independent Domination in Regular-DIM Graphs

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Abstract

Let F be an indexed CNF on k variables. Call a partial assignment α bilateral when every unassigned variable occurs in both polarities among the clauses not already satisfied by an assigned literal, and define

$$\beta(F) = \min_{\alpha \text{ bilateral}} (|T_F(\alpha)| - |U(\alpha)|).$$

We prove that β is the minimum ordinary deficiency of a bipolar residual restriction. A size-preserving bijection with independent dominating sets of the clause-literal formula graph $G(F)$ gives $i(G(F)) = k + \beta(F)$ and a generating-polynomial identity. The parameter is additive, invariant under width lifting, equal to the MaxSAT defect at width at most two, and sufficient to recover MaxSAT after clause replication. With unrestricted occurrences, its sign is polynomial-time decidable at width two and NP-complete for every fixed uniform width $r \geq 3$; the proper, simple, exact signed-occurrence $(3, 2, 2)$ slice remains open.

Every regular graph equipped with a dominating induced matching is coordinatised by a formula satisfying $i(G) - \mu^*(G) = \beta(F)$. Conditional expectation gives the constructive bound

$$i(G) \leq \mu^*(G) + \lfloor \frac{2(d-1)}{d \cdot 2^d} \mu^*(G) \rfloor$$

in the d -regular DIM class. A published enforcer yields a connected amplifier with $\beta(A(R)) = \varrho(R)$, connected linear-gap families, and asymptotic gap density $1/72$. An imported 20-clause lower bound proves that order 50 is minimum among cubic graphs admitting a dominating induced matching and satisfying $i(G) > \mu^*(G)$. Three finite encodings have native- and Lean-checked LRAT proofs. Those checks do not formalise the universal theory; the release is unrefereed and has no independent reproduction or specialist review.

Keywords: CNF deficiency, partial assignment, independent domination, dominating induced matching, MaxSAT, parameterized complexity, formula graph

設 F 為含 k 個變數的具索引 CNF。若每個未指派變數在剩餘子句中 均保留正、負兩種極性，則該部分指派稱為雙側，並定義雙側虧缺 $\beta(F) = \min(|T_F(\alpha)| - |U(\alpha)|)$ 。本文證明它是雙極剩餘限制的最小普通虧缺；雙側指派與子句-文字圖的獨立支配集存在保持大小的雙射，故 $i(G(F)) = k + \beta(F)$ ，生成多項式亦相應。參數可加、在寬度提升下不變；寬度至多二時等於 MaxSAT 缺陷。若不限制文字出現次數，其符號判定在寬度二屬 P，而在每個固定均勻子句寬度 $r \geq 3$ 為 NP 完全；精確符號出現次數 $(3, 2, 2)$ 的情形仍未分類。對附有支配誘導匹配的正則圖， $i(G) - \mu^*(G) = \beta(F)$ ，並有可建構上界。發表的強制元件給出值等於 骨架最小邊覆蓋的連通放大器；O--West 極值骨架使漸近差距密度達 $1/72$ 。外部二十子句下界證明：在承認支配誘導匹配的三次圖中，違反不等式的最小階數為 50。三組 SAT/LRAT 有限命題由原生檢查器與 Lean 檢查；理論仍是未經同儕審查的書面證明，尚無獨立重現或完整形式化。

關鍵詞：CNF 虧缺；部分指派；獨立支配；支配誘導匹配；MaxSAT；參數化複雜度

1. Introduction

A graph-theoretic construction can expose a useful expression without establishing a new formula parameter. This distinction matters in the setting that motivated the present work. A verified release resolving the third TxGraffiti conjecture used a 15-variable exact signed-occurrence $(3, 2, 2)$ -CNF to construct a cubic graph with a dominating

induced matching and with independent domination number one larger than its minimum maximal-matching number [1]. The formula side of the construction naturally produced a residual objective: the number of clauses left unsatisfied by the assigned literals, minus the number of variables deliberately left unassigned. The immediate question was whether this expression merely recorded the accounting of that finite graph, or whether it supported a formula-side theory independent of the example.

The standard for a positive answer should be demanding. A distinct optimisation parameter ought to have a specified domain on general indexed formulas, an intrinsic interpretation, exact translations to neighbouring problems, separation from established quantities, nontrivial algebra, a computational theory, and uses that survive beyond the motivating instance. It should also be described within a defensible priority boundary. The last condition is particularly important here because the closest prior reduction, due to Chlebík and Chlebíková, already expresses the size of an independent dominating set as a literal-selection cost plus a penalty for bad clauses [2]. The new theory therefore cannot be presented as the first appearance of the underlying arithmetic.

This paper develops the resulting parameter, called **bilateral deficiency**. The adjective records its feasible-domain condition: every unassigned variable must retain both literal polarities in the residual. The principal structural result is a size-preserving bijection between bilateral assignments of an indexed CNF F and independent dominating sets of a canonical formula graph $G(F)$. The objective then becomes an exact coordinate:

$$i(G(F)) = |V(F)| + \beta(F).$$

The bijection preserves not only the optimum but every solution size and its multiplicity. On the formula side, β is precisely the least clause-variable deficiency among residual restrictions that remain bipolar. These two characterisations give the parameter both a SAT semantics and a graph semantics.

Four further groups of results establish that the definition has mathematical life beyond this identity. First, explicit examples separate it from ordinary deficiency, maximum deficiency, satisfiability, and MaxSAT defect. Second, it is additive on variable-disjoint unions, its complete solution polynomial is multiplicative, and a local width-lifting operation preserves its value. Third, its complexity has a sharp width boundary: at width at most two it equals the MaxSAT defect, while the sign problem is NP-complete on proper simple r -uniform CNF for every fixed $r \geq 3$. Indexed clause replication recovers MaxSAT exactly, and bounded incidence treewidth gives a single-exponential parameterized algorithm. Fourth, regular graphs equipped with a dominating induced matching admit a complete coordinatisation by a signed-occurrence formula class. Within that class, β is exactly the signed gap $i - \mu^*$, and a constructive probabilistic argument bounds this gap above by $2(d-1)\mu^*/(d2^d)$. A parser-independent Lean-checked finite terminal signature for the published $E_{3,2,2}$ enforcer then supports a connected composition theorem: attaching one enforcer to every edge of a connected cubic skeleton R makes β equal to the minimum edge-cover number of R . Prisms give an explicit infinite family with a linear positive gap.

These theorems support six concrete uses. They give a necessary-and-sufficient search objective for counterexamples within formula-generated regular-DIM graphs, concise upper witnesses and compositional lower proofs, structural algorithm routing, prescribed-value benchmark generation, a full sign classification of a structured graph family, and a typed claim-evidence interface for AI-assisted mathematics. Each use has a qualification. Systematic discovery is not polynomial-time discovery, lower-bound proofs need not be short, theorem-level routing does not predict the fastest engineering implementation, and technical infrastructure does not demonstrate community adoption.

The paper proceeds by fixing the prior-art boundary before introducing the definition. Section 3 proves the residual and graph characterisations. Section 4 establishes separation and algebra. Section 5 develops complexity and algorithm selection. Section 6 coordinatises the regular-DIM family. Section 7 evaluates the six uses, and Section 8 records reproducibility limits and open problems. All general statements are supported by written proofs. Computation is used only to check finite objects and implementations, not as a substitute for the universal arguments.

2. Prior art and claim boundary

2.1 Formula graphs and independent domination

Several SAT-to-graph constructions use a complementary pair of literal vertices for each variable and clause vertices adjacent to their literals. What changes between constructions is the clause-side adjacency and the optimization objective. Zverovich's satgraph makes the clause vertices a clique [3]. At most one of those vertices can then appear in an independent set, and the independent domination optimum collapses to a satisfiability indicator: k for a satisfiable k -variable formula and $k+1$ for an unsatisfiable one. That construction establishes a close architectural antecedent, but it deliberately discards the number and structure of residual clauses.

Clause-literal encodings also serve exact formula generation. Zhang, Peitl, and Szeider used such representations to generate and classify small unsatisfiable k -CNFs under literal-occurrence bounds [4]. Their work is especially relevant to the exact signed-occurrence $(3, 2, 2)$ class used later in this paper: each clause has width three and each signed literal occurs twice. Ahadi and Dehghan proved hardness results for the corresponding twice-positive, twice-negative form of 3-SAT and applied that source problem to domination questions [5]. These papers establish that exact signed-occurrence spaces are neither artificial nor computationally trivial. They optimize satisfiability or a target graph problem, however, not the residual difference defined here.

2.2 The closest antecedent: the replicated-clause cost

The strongest overlap occurs in Chlebík and Chlebíková's approximation reduction for bounded-degree Minimum Independent Dominating Set [2]. The exact pinpoint is Section 2.3, proof of Theorem 5, pp. 10--11 of the authors' preprint. Their graph contains two adjacent literal vertices for each variable and t indexed vertices for each clause. An independent dominating set D has a literal part D_1 , which determines a partial assignment. Clauses without a selected literal are called bad. If the formula has $5k$ clauses and a fraction b is bad, the paper puts all t copies of every bad clause into D_2 and records

$$|D| = |D_1| + |D_2| = |D_1| + (5k)bt.$$

Equivalently, with T the indexed set of bad clauses, the decomposition is $|D| = |D_1| + t|T|$.

Writing n for the number of formula variables and letting U denote variables for which neither literal was selected, $|D_1| = n - |U|$. The generic replicated cost is therefore $n + t|T| - |U|$. At $t = 1$, this is exactly the objective used below, shifted by the constant n . In the notation of [2], $n = 3k$; the neutral letter prevents their scaling parameter from being confused with the variable count used elsewhere in this paper.

This antecedent decides the novelty boundary. The decomposition itself is not new. What is not developed in [2] is the exact formula-side feasible domain forced by domination of both endpoints of every unselected literal pair; a converse mapping for all feasible partial assignments; a residual clause-variable parameter; an optimum identity and full solution polynomial; or the algebraic, complexity, and regular-graph classification results proved here. The relationship is therefore one of extraction and extension: an implicit cost in an approximation reduction becomes an explicit residual optimization object with its own theory.

2.3 Deficiency parameters in SAT

For a formula with m indexed clauses and n variables, ordinary deficiency is $m - n$. SAT theory also studies maximum deficiency over clause subfamilies, matching autarkies, lean kernels, and surplus. Kullmann gives a systematic clausal theory connecting autarkies and deficiency [6], while Szeider analyzes bounded clause-variable difference in minimally unsatisfiable formulas and parameterized satisfiability [7]. These quantities are close enough that a new definition must be distinguished by its optimization domain, not by a renamed arithmetic expression.

The relevant quantities differ as follows.

Quantity	Objects optimized over	Objective	Feasibility condition
ordinary deficiency	the presented indexed formula	$ F - V(F) $	none
maximum deficiency / surplus	clause subfamilies or their variable neighborhoods	maximum clause--variable excess, or its dual	subformula or neighborhood choice
autarky / lean-kernel methods	partial assignments and clauses they touch	remove or characterize autarkic structure	every touched clause is satisfied
MaxSAT defect τ	complete assignments	falsified indexed clauses	every variable assigned
bilateral deficiency β	residual restrictions induced by partial assignments	$ T - U $	every unassigned variable survives in both signs
replicated-clause cost of [2]	literal selections in one reduction graph	$ D_1 + t T $	graph independence and domination

Bilateral deficiency minimizes, rather than maximizes, a clause-variable difference. More importantly, it minimizes over partial-assignment residuals subject to a two-polarity survival condition. It does not range over arbitrary clause subfamilies, and a feasible bipolar residual need not be lean. Ordinary deficiency evaluates one presented formula. Maximum deficiency scans subformulas. Bilateral deficiency scans residual restrictions. The separation examples in Section 4 prove that neither ordinary nor maximum deficiency determines its value.

The polarity condition also prevents an identification with an unrestricted residual minimum. Without it, leaving variables unassigned would earn a unit reward even when those variables had disappeared from every surviving clause. The optimum would then depend on the declared variable universe in an unstructured way. Bilaterality makes each reward accountable: an unassigned variable must still participate on both sides of the remaining constraint system. This is analogous in spirit to matching conditions that organize autarky theory, but its quantifiers and objective differ. An autarky satisfies every clause it touches. A bilateral assignment constrains the variables it does not assign. A lean kernel removes autarkic structure. A bilateral residual is selected by minimizing its own deficiency and need not be lean. Existing deficiency language is therefore essential context without making the new optimum redundant.

2.4 Priority statement

The literature inspection supports the following calibrated statement. Bilateral deficiency appears to be new as an explicitly defined SAT-style residual optimization parameter together with the bijective, algebraic, complexity, and regular-DIM theory proved here. The partial-assignment cost decomposition is implicit in Chlebík and Chlebíková [2]. An earlier source for the independent domination polynomial was also located and is credited below [8]. The dated search log accompanying the artifact records exact-phrases and concept searches over SAT deficiency, residual restrictions, autarkies, formula graphs, independent domination, domination polynomials, and the 2008 reduction lineage.

This is not a claim of global priority. The inspection used public web and publisher indexes and targeted citation chaining, but did not exhaust MathSciNet, zbMATH, dissertations, non-English literature, hypergraph-transversal terminology, or all non-digitized sources. The mathematical results below do not depend on priority, and a subject specialist would need to re-audit the historical description before any future journal submission.

2.5 Imported theorem dependencies

Three external mathematical inputs are load-bearing. The table records the exact imported object or result, the convention transfer used here, and the consequence checked locally. This separates attribution and interface checks from the original proofs in their cited sources.

Imported source	Exact input used	Convention transfer	Locally checked consequence
Zhang--Peitl--Szeider [4, Appendix A.1]	The displayed ten-clause $E_{3,2,2}$ enforcer with one terminal forced false in every satisfying assignment	Variables and clauses are transcribed into the present indexed-CNF convention; a global polarity switch fixes the terminal orientation; internal signed occurrences and the terminal's two negative occurrences are recounted	The hard-coded formula and its four-entry terminal signature are checked independently of the production DIMACS parser; the signature then enters Theorem 12
Zhang--Peitl--Szeider [4, Theorem 15 and Table 1]	The smallest unsatisfiable $(3, 2, 2)$ -formula has 20 distinct clauses	Tautologies are deleted and duplicate indexed clauses collapsed; clause width is at most three and each signed literal occurs at most twice, matching the source's bounded-occurrence convention	Positive β implies unsatisfiability; the incidence identities $3m = 4k$ and $ V(G) = 2k + m$ then give the order-50 lower bound in Corollary 13
O--West [13, Corollaries 2.3 and 4.4; Theorem 5.2]	Every connected cubic graph of order r has matching number at least $(4r - 1)/9$, with an infinite equality family H_1	Their n is the present skeleton order r , and their matching number α' is the present v	Theorem 12 converts $r - v(R)$ into the amplifier gap; the displayed family parameters are checked before deriving the construction-specific density $1/72$

The local checks validate transcription, syntax, and the stated consequences. They do not independently re-prove the imported theorems.

3. Bilateral residuals and the formula-graph bijection

3.1 Indexed residual semantics

An **indexed CNF** is a finite sequence $F = (C_a)_{a \in A}$ of clauses on an explicit finite variable set $V(F) = \{x_1, \dots, x_k\}$. A clause is a set of literals: repeated positions within a clause collapse, while extensionally equal clauses at different indices remain distinct. This distinction is essential because clause replication is one of the algebraic operations. Empty and tautological clauses are permitted unless a restricted formula class excludes them. An explicit variable may occur in no clause.

A partial assignment is a map $\alpha : V(F) \rightarrow \{0, 1, *\}$, where $*$ denotes unassigned. Define

$$U(\alpha) = \{x \in V(F) : \alpha(x) = *\}$$

and let $T_F(\alpha) \subseteq A$ contain exactly those clause indices for which no literal has been made true by α . The residual $F \upharpoonright \alpha$ is obtained by deleting satisfied indexed clauses and deleting assigned false literals from every survivor. It retains the original indices. A residual is **bipolar** if each variable occurring in it occurs with both signs.

The assignment α is **bilateral** if every variable in $U(\alpha)$ occurs positively and negatively among the clauses indexed by $T_F(\alpha)$. This requirement does two jobs. It ensures that each explicit unassigned variable actually survives, and it makes the residual bipolar. Complete assignments are bilateral vacuously. Thus the feasible set is always nonempty, even when F contains empty clauses or unused variables.

Define the residual objective and its optimum by

$$\delta_F(\alpha) = |T_F(\alpha)| - |U(\alpha)|, \quad \beta(F) = \min_{\alpha \text{ bilateral}} \delta_F(\alpha).$$

The use of indexed clauses means that $|T_F(\alpha)|$ counts multiplicity. The use of an explicit variable set means that assigning an otherwise absent variable is sometimes necessary for feasibility. These conventions remove two ambiguities that would otherwise make replication and graph translation ill-defined.

3.2 Least deficiency of a bipolar residual

Theorem 1 (residual-core characterization). For every indexed CNF F ,

$$\beta(F) = \min_{\substack{\alpha: F \upharpoonright \alpha \text{ bipolar} \\ V(F \upharpoonright \alpha) = U(\alpha)}} (c(F \upharpoonright \alpha) - n(F \upharpoonright \alpha)).$$

Proof. The surviving indexed clauses are exactly those in $T_F(\alpha)$, so their number is $|T_F(\alpha)|$. The assignment is bilateral precisely when every unassigned variable occurs in both polarities in the residual. Under that condition the residual is bipolar and its variable set is exactly $U(\alpha)$. Its ordinary deficiency is therefore $|T_F(\alpha)| - |U(\alpha)|$. Conversely, every residual appearing in the displayed domain satisfies the bilateral condition. The two minima range over the same assignments with the same objective. $\square$

The theorem provides the requested intrinsic SAT interpretation. The parameter is not defined by reference to a graph, although the graph translation will make it computationally and structurally useful. It selects the least deficient bipolar core obtainable by restriction, with explicit control of variables that have disappeared.

3.3 A size-preserving bijection

Construct the **formula graph** $G(F)$ as follows. For each variable x_j , create adjacent literal vertices v_j^+ and v_j^- . For every indexed clause C_a , create a clause vertex c_a . Clause vertices are mutually nonadjacent, and c_a is adjacent to the literal vertex for each literal in C_a . Tautological clauses are adjacent to both endpoints of the relevant pair. Equal clauses at different indices give distinct twin clause vertices.

Theorem 2 (bilateral assignment--independent domination bijection). The map

$$\alpha \mapsto D_\alpha = \{\text{the true literal vertex selected for every assigned variable}\} \cup \{c_a : a \in T_F(\alpha)\}$$

is a bijection from the bilateral assignments of F to the independent dominating sets of $G(F)$. Moreover,

$$|D_\alpha| = k + \delta_F(\alpha), \quad i(G(F)) = k + \beta(F).$$

Proof. Let α be bilateral. At most one endpoint of each literal pair is selected. A selected clause vertex has no selected literal neighbor, because its clause has no assigned true literal. Clause vertices are mutually nonadjacent, so D_α is independent.

Every assigned literal pair is dominated by its selected endpoint. If a clause is not in $T_F(\alpha)$, an assigned true literal dominates it; if it is in $T_F(\alpha)$, its own clause vertex is selected. For an unassigned variable, bilaterality supplies a selected residual clause adjacent to the positive endpoint and a selected residual clause adjacent to the negative endpoint. The two clauses may coincide when a tautology is present. Hence D_α dominates all vertices. Its size is

$$(k - |U(\alpha)|) + |T_F(\alpha)| = k + \delta_F(\alpha).$$

Conversely, let D be an independent dominating set. Independence allows at most one endpoint from each literal pair. Assign the corresponding truth value when one endpoint is selected and leave the variable unassigned when neither is selected. A clause vertex belongs to D exactly when it has no selected literal neighbor: independence proves that a selected clause has no such neighbor, and domination forces every unselected clause to have one. Thus the selected clause vertices are precisely $T_F(\alpha_D)$. If neither endpoint of a literal pair is selected, both endpoints must be dominated by selected clause vertices. The residual consequently contains both polarities of that variable, so α_D is bilateral. Both constructions recover their inputs, and the size equation already proved completes the result. $\square$

The converse is the step that turns a one-way reduction cost into a genuine parameter identity. It shows that no independent dominating set falls outside the formula-side feasible domain and no bilateral assignment is lost.

Several boundary cases show why the indexed formulation is not cosmetic. An empty clause has an isolated clause vertex and must be selected in every independent dominating set; on the formula side it survives every assignment and contributes one to $|T|$. A tautological clause is always satisfied by a complete assignment, but under an unassigned variable it can dominate both literal endpoints from one selected clause vertex. The bilateral definition therefore permits the same residual clause to witness both polarities. Duplicate indexed clauses become distinct independent twins and are counted separately on both sides. An explicit variable absent from every clause cannot be left unassigned by a bilateral assignment, just as neither endpoint of its isolated complementary edge can be dominated by a clause vertex. These cases agree without auxiliary exceptions.

The bijection is compatible with witnesses as well as values. A formula-side certificate records a ternary vector and its residual clause indices. Its image is obtained without search. A graph-side independent dominating set recovers the ternary vector by inspecting complementary pairs, and its selected clause vertices must equal the recomputed residual indices. This round trip gives a linear-time checker once the formula and candidate are fixed. It does not certify optimality, but it makes every claimed upper bound transparent.

3.4 Complete solution polynomial

Define a Laurent polynomial on the formula side by

$$B_F(z) = \sum_{\alpha \text{ bilateral}} z^{\delta_F(\alpha)}.$$

Let $D_i(G, z) = \sum_D z^{|D|}$, where the sum ranges over all independent dominating sets. This polynomial was introduced under the name independent domination polynomial by Dod [8] and subsequently studied by Jahari and Alikhani [9]. Theorem 2 immediately gives more than equality of optima.

Corollary 3. For every indexed CNF F on k variables,

$$D_i(G(F), z) = z^k B_F(z).$$

Every exponent and coefficient is preserved after the constant shift. Thus the formula parameter carries the full independent-domination solution-size spectrum on formula graphs, including multiplicities. This identity provides a strong cross-model test: independently enumerating bilateral assignments and graph independent dominating sets must yield the same polynomial, not merely the same minimum.

3.5 Replicated clauses and the 2008 construction

For an integer $t \geq 1$, define

$$\beta_t(F) = \min_{\alpha \text{ bilateral}} (t|T_F(\alpha)| - |U(\alpha)|).$$

Let $F^{[t]}$ repeat every indexed clause t times. Replication preserves the bilateral feasible domain and multiplies the surviving-clause count, so

$$\beta_t(F) = \beta(F^{[t]}), \quad i(G(F^{[t]})) = k + \beta_t(F).$$

The graph used in [2] is therefore the replicated-clause member of this exact framework. This observation credits the antecedent while clarifying what the parameter adds: the domain, inverse map, residual semantics, and theory remain visible for every replication weight.

4. Separation and algebra

4.1 Separation from standard SAT quantities

The arithmetic $|T| - |U|$ resembles deficiency, but its optimum is not determined by ordinary or maximum deficiency. Consider

$$F_0 = (x) \wedge (y) \wedge (x \vee y), \quad F_1 = (x) \wedge (y) \wedge (\neg x \vee \neg y).$$

Both formulas have two variables and three clauses, so ordinary deficiency is one. Every proper clause subfamily has deficiency at most one, so their maximum deficiencies also agree. Formula F_0 is satisfiable and, by the width-two theorem below, has $\beta(F_0) = 0$. Formula F_1 cannot satisfy all three clauses but can satisfy two, hence $\beta(F_1) = 1$. Equal ordinary and maximum deficiencies can therefore coexist with different bilateral deficiencies.

The comparison is stronger than a sign example. Both formulas have the same numbers of variables and indexed clauses, the same clause-width profile $(1, 1, 2)$, and the same maximum deficiency. Their different values arise from polarity geometry under restriction. In F_0 , a satisfying completion reaches zero. In F_1 , every completion leaves one clause false, and the width-two theorem shows that no strategically unassigned variable can improve the objective. Bilateral deficiency consequently distinguishes formulas that all of these coarser statistics identify.

Nor is β a relabelled satisfiability or MaxSAT answer. Let

$$H = (x \vee y \vee z) \wedge (\neg x \vee \neg y \vee \neg z).$$

The empty assignment is bilateral and has deficiency $2 - 3 = -1$, while H is satisfiable and has MaxSAT defect zero. The width-three occurrence bound $2|U| \leq 3|T|$ excludes any value below -1 , so $\beta(H) = -1$. Conversely, Proposition 14 proves that the explicit 15-variable exact signed-occurrence $(3, 2, 2)$ component P is unsatisfiable and has value $+1$. Combining that positive component with the negative component of Lemma 8 by disjoint union gives unsatisfiable formulas of value zero and negative one. Satisfiability therefore does not determine even the sign of β .

4.2 Additivity and spectral convolution

Theorem 4 (disjoint-union algebra). If F_1 and F_2 have disjoint variable sets, then

$$\beta(F_1 \sqcup F_2) = \beta(F_1) + \beta(F_2), \quad B_{F_1 \sqcup F_2}(z) = B_{F_1}(z)B_{F_2}(z).$$

Proof. Every partial assignment on the union factors uniquely into a pair of partial assignments. It is bilateral exactly when both restrictions are bilateral, because no literal or clause crosses the variable partition. Surviving clause counts and unassigned variable counts add. Hence objectives add for every feasible pair. Taking minima proves the first identity, while summing the monomials over all feasible pairs proves the polynomial product. $\square$

This algebra is nontrivial in two senses. It controls optimum values and full solution distributions, and it supports compositional certificates. Once the value of each distinct component has been proved, arbitrarily large block instances can be checked by verifying the partition and adding component values. The operation also shows that negative values are not pathological one-off events: any positive and negative base components generate a two-sided integer semigroup.

The polynomial identity retains information discarded by the minimum. If $b_j(F)$ is the number of bilateral assignments of deficiency j , then the coefficient rule

$$b_j(F_1 \sqcup F_2) = \sum_{p+q=j} b_p(F_1)b_q(F_2)$$

is an exact convolution. It permits distribution-sensitive tests in which two formulas have equal β but different near-optimal spectra. On the graph side, the same coefficients count independent dominating sets at each size after the variable shift. Additivity is therefore not merely a device for manufacturing optimum values. It transports a complete counting invariant through disjoint composition.

4.3 Exact width lift

For every clause C_a , introduce a fresh variable y_a and replace the clause by

$$C_a \vee y_a, \quad C_a \vee \neg y_a.$$

Write $L(F)$ for the lifted formula.

Theorem 5 (width-lift invariance). For every indexed CNF F ,

$$\beta(L(F)) = \beta(F).$$

If F is a proper, duplicate-free r -uniform CNF, then $L(F)$ is a proper, duplicate-free $(r + 1)$ -uniform CNF.

Proof. Restrict a partial assignment on $L(F)$ to the original variables. If C_a is satisfied by that restriction, both lifted clauses disappear. Bilaterality then forces y_a to be assigned, since an unassigned y_a would have neither polarity in the residual. The original clause and its lifted pair both contribute zero to the objective.

Suppose C_a survives. If y_a is assigned, exactly one lifted clause survives; if y_a is unassigned, both survive and the new variable adds one to $|U|$. In either case the net contribution is one, exactly the contribution of the original surviving clause. The surviving lifted clauses contain the same unassigned original literals, so bilaterality of every original variable is preserved in both directions. Local choices for distinct y_a are independent. The feasible objectives therefore correspond exactly, proving equality of minima. Fresh variables distinguish the two children of every clause and preserve properness and uniform clause width. $\square$

Repeated lifting moves a construction from width two to every larger uniform clause width without changing β . This makes the operation useful both in hardness reductions and in algebraic conformance benchmark design.

4.4 Recovering MaxSAT by clause replication

Let $\tau(F)$ denote the minimum number of clauses falsified by a complete assignment. Every completion of a partial assignment can falsify only clauses that survived the partial assignment, so $|T_F(\alpha)| \geq \tau(F)$.

Theorem 6 (MaxSAT recovery). If F has k variables and $r > k$, then

$$\tau(F) = \lceil \frac{\beta(F^{[r]})}{r} \rceil.$$

More precisely,

$$\beta(F^{[r]}) = r\tau(F) - u^*,$$

where u^* is the maximum number of variables left unassigned by a bilateral assignment with exactly $\tau(F)$ original residual clauses.

Proof. Replication changes the objective to $r|T| - |U|$. Since $|T| \geq \tau(F)$ and $|U| \leq k$, every feasible assignment has value at least $r\tau(F) - k$. A complete MaxSAT-optimal assignment is bilateral and has value $r\tau(F)$. Thus

$$r\tau(F) - k \leq \beta(F^{[r]}) \leq r\tau(F).$$

Because $r > k$, dividing by r and taking the ceiling yields $\tau(F)$. A minimizer cannot have $|T| \geq \tau(F) + 1$, since then its value is at least $r(\tau(F) + 1) - k > r\tau(F)$. It must therefore have exactly $\tau(F)$ residual clauses and maximize $|U|$, proving the refined formula. $\square$

The result places β in a precise relation to MaxSAT. At unit weight it can differ sharply from the MaxSAT defect. Under a sufficiently large indexed penalty it contains both the defect and a secondary optimum measuring how many variables can remain bilateral at MaxSAT optimality.

5. Complexity and algorithm selection

5.1 Baseline decision problem

Define **BILATERAL-DEFICIENCY** as the decision problem asking whether $\beta(F) \leq q$. It belongs to NP: a ternary assignment is a polynomial-size certificate, and one can scan the indexed clauses to check the residual count, the unassigned count, and both-polarity occurrence for every unassigned variable. Direct exact enumeration takes

$O(3^k |F|)$ time and polynomial space. This baseline is useful for small k , but the next results show that syntax should guide a solver whenever additional structure is present.

The input conventions affect this membership statement only in predictable ways. Clause indices and the explicit variable count are part of the instance, so duplicates and unused variables require no reconstruction. For each surviving clause, a verifier records the signs of its unassigned literals only after confirming that no assigned literal satisfies the clause. It then tests two sign bits per unassigned variable. The certificate checker is linear in the total literal-incidence size, apart from integer arithmetic for the threshold. This ordering matters in implementations: recording a polarity from a clause that is later discovered to be satisfied creates a false bilateral witness.

The definition also has a direct 0–1 optimization model. All variables in the following formulation are explicitly binary. For each formula variable x_j , introduce $a_j^0, a_j^1, u_j \in \{0, 1\}$ with $a_j^0 + a_j^1 + u_j = 1$. For each indexed clause a , let r_a indicate that no assigned true literal occurs in that clause, with $r_a \in \{0, 1\}$. If s_ℓ denotes a_j^1 for a positive literal x_j and a_j^0 for a negative literal $\neg x_j$, impose

$$r_a \leq 1 - s_\ell \quad (\ell \in C_a), \quad r_a \geq 1 - \sum_{\ell \in C_a} s_\ell.$$

These constraints make r_a exactly the residual-clause indicator, including for empty and tautological clauses. Bilaterality is expressed by

$$u_j \leq \sum_{a: x_j \in C_a} r_a, \quad u_j \leq \sum_{a: \neg x_j \in C_a} r_a.$$

Minimizing $\sum_a r_a - \sum_j u_j$ is therefore an exact binary linear formulation of $\beta(F)$. It offers a general MILP route and, more importantly, a second executable specification against which specialized solvers can be checked. A solver optimum still requires a verifiable lower bound or trusted optimization proof. The formulation alone does not turn an optimizer's status line into a certificate.

5.2 Width at most two

Theorem 7 (width-two equality). If every clause of F has width at most two, then

$$\beta(F) = \tau(F).$$

The statement includes indexed duplicate clauses, units, empty clauses, and tautological binary clauses.

Proof. A complete MaxSAT-optimal assignment is bilateral, so $\beta(F) \leq \tau(F)$. For the reverse inequality, take any bilateral assignment whose residual has u variables and t indexed clauses. Form the **signed literal-occurrence bipartite multigraph** of the residual. Its left vertices are residual variables, its right vertices are indexed residual clauses, and each signed literal occurrence is a separate edge. Thus a tautological binary clause containing x and $\neg x$ gives two parallel edges from variable x to one clause vertex. For a set X of residual variables, let $N(X)$ be the underlying set of adjacent clause vertices. Bilaterality supplies at least two signed occurrences per variable in X . All such incidences end in $N(X)$, and each residual clause has at most two signed literal occurrences. Consequently

$$2|X| \leq e(X, N(X)) \leq 2|N(X)|.$$

Hall's condition holds, so every residual variable can be matched to a distinct containing clause. Assign each variable to satisfy its literal in the matched clause. The matched clauses are distinct, hence at least u of the t residual clauses become satisfied. Completing any remaining choices therefore falsifies at most $t - u$ clauses. It follows that $\tau(F) \leq t - u$ for every bilateral assignment. Minimization gives $\tau(F) \leq \beta(F)$, completing the equality. $\square$

The proof explains why width two is the exact boundary for this argument. Bilaterality supplies two incidences per residual variable. Width two gives at most two incidences per residual clause, so Hall's inequality follows with coefficient one. At width three, the same count yields only $2|X| \leq 3|N(X)|$, which cannot match all variables to distinct clauses. Lemma 8 exploits precisely this slack. The complexity dichotomy below is thus visible in the combinatorial proof rather than appearing as an unrelated reduction artifact.

Empty clauses do not disturb the matching argument because they consume no variable incidence and may remain among the $t - u$ falsified clauses. A unit clause contributes capacity one rather than two, which only strengthens the upper incidence bound. A tautological binary clause contains both literals of one variable but remains a single distinct clause vertex, and the argument still uses only its two incidences.

The equality has two different algorithmic consequences. First, $\beta(F) = 0$ exactly when a width-two formula is satisfiable, so the sign is decided by 2-SAT in polynomial time. Second, arbitrary thresholds encode the clause-deletion or Almost-2-SAT objective. The latter is fixed-parameter tractable in the permitted number q of falsified

clauses. Razgon and O'Sullivan gave an $O(15^q m^3)$ algorithm [10]. Thus a restricted slice can have an easy sign test and an NP-hard variable threshold without contradiction.

5.3 NP-complete thresholds already at uniform width two

The threshold hardness has a direct graph reduction. Given a simple graph $H = (V, E)$, create a variable x_v for every vertex. For each edge uv , include the two clauses

$$(x_u \vee x_v), \quad (\neg x_u \vee \neg x_v).$$

Under a complete assignment, a cut edge falsifies neither clause and an uncut edge falsifies exactly one. Theorem 7 yields

$$\beta(F_H) = \tau(F_H) = |E| - \text{MaxCut}(H).$$

Simple MaxCut is NP-complete [11]. Therefore BILATERAL-DEFICIENCY is NP-complete even for proper duplicate-free 2-uniform CNF when q is part of the input.

5.4 A negative exact signed-occurrence (3, 2, 2) component

The fixed-threshold hardness proof needs a proper uniform-width component with negative value. Let N be

$$\begin{aligned} & (x_4 \vee x_5 \vee \neg x_6) \wedge (\neg x_1 \vee x_3 \vee \neg x_6) \wedge (\neg x_2 \vee x_3 \vee x_4) \\ & \wedge (\neg x_1 \vee \neg x_4 \vee \neg x_5) \wedge (x_2 \vee \neg x_5 \vee x_6) \wedge (x_1 \vee \neg x_2 \vee \neg x_3) \\ & \wedge (x_2 \vee \neg x_3 \vee \neg x_4) \wedge (x_1 \vee x_5 \vee x_6). \end{aligned}$$

Every clause contains three different variables, no clauses repeat, and each signed literal occurs exactly twice.

Lemma 8. $\beta(N) = -1$.

Proof. Set $x_1 = 1$ and $x_5 = x_6 = 0$, leaving x_2, x_3, x_4 unassigned. Exactly two clauses survive:

$$(\neg x_2 \vee x_3 \vee x_4), \quad (x_2 \vee \neg x_3 \vee \neg x_4).$$

Each unassigned variable occurs with both signs, so the residual is bilateral and has value $2 - 3 = -1$. For a lower bound, every width-three bilateral residual with u variables and t clauses has at least $2u$ literal occurrences and at most $3t$, hence $2u \leq 3t$. If $t - u \leq -2$, then $t \leq u - 2$, and the two inequalities imply $u \geq 6$. Formula N has only six variables, so $u = 6$, the assignment is empty, and all eight clauses survive. This contradicts $t \leq 4$. No smaller value exists. $\square$

The proof illustrates a useful certificate pattern. An upper bound is one partial assignment. The lower bound is a global occurrence inequality plus a finite boundary case, not a trust in solver output.

5.5 Fixed-threshold hardness from width three onward

Theorem 9 (sign dichotomy by width). For every fixed $r \geq 3$, deciding $\beta(F) \leq 0$ is NP-complete on proper simple r -uniform CNF. Its complement, deciding $\beta(F) > 0$, is coNP-complete on the same class.

Proof. Reduce from the Simple MaxCut decision problem: given H and K , decide whether H has a cut of size at least K [11]. We restrict without loss of generality to $1 \leq K \leq m = |E(H)|$. An input with $K \leq 0$ is trivially positive and is replaced by the fixed one-edge instance with threshold one; an input with $K > m$ is trivially negative and is replaced by the fixed triangle instance with threshold three. Both replacements satisfy $1 \leq K \leq m$, preserve the answer, and ensure $m \geq 1$. Form the 2-uniform CNF F_H above. Then $\beta(F_H) = m - \text{MaxCut}(H)$. Apply one width lift, preserving the value and producing a proper simple 3-uniform CNF. Add $m - K$ variable-disjoint copies of N . By Theorem 4 and Lemma 8, the resulting formula F' satisfies

$$\beta(F') = m - \text{MaxCut}(H) - (m - K) = K - \text{MaxCut}(H).$$

Thus $\beta(F') \leq 0$ exactly when the MaxCut instance is positive. The reduction is polynomial and the problem is in NP. Repeated width lifts prove the statement for each fixed $r > 3$. Complementation gives the coNP result. $\square$

The problem parameterized only by q is consequently para-NP-hard: the fixed slice $q = 0$ remains NP-hard at uniform width three. This rules out an FPT algorithm in q alone unless $P = NP$, contrasting with Almost-2-SAT. The complexity of the sign problem on the narrower proper exact signed-occurrence (3, 2, 2) class remains open.

Satisfiability hardness for that class [5] does not resolve the question because Section 4 showed that satisfiability does not determine the sign of β .

5.6 Bounded incidence treewidth

Let $I(F)$ be the unsigned variable-clause incidence graph. Suppose it has a tree decomposition of width t . Replace every variable in each bag by its two literal vertices and retain the clause vertices. The resulting bags cover all formula-graph incidence edges and all complementary-pair edges, and they preserve connected bag occurrence for every vertex. Their size is at most twice the original bag size, giving formula-graph treewidth at most $2t + 1$.

Independent dominating sets are generalized (σ, Q) -sets with $\sigma = \{0\}$ and $Q = \{1, 2, \dots\}$. The bounded-treewidth algorithms of Focke et al. apply to finite or cofinite degree sets and count sets of a specified size [12]. With a decomposition supplied, one invocation for each size $s = 0, \dots, |V(G(F))|$ yields every coefficient of the independent domination polynomial. This polynomial number of calls preserves the bound $2^{O(t)}|F|^{O(1)}$. Each coefficient is at most $2^{|V(G(F))|}$, so its binary representation has $O(|F|)$ bits and the complete explicit output has polynomial bit length. Taking the least nonzero size also gives the optimum. Through Theorem 2 and Corollary 3, this computes $\beta(F)$ and the complete Laurent polynomial $B_F(z)$. This is an exact structural route, not a heuristic association between incidence width and solver performance.

5.7 Theorem-driven solver routing

The complexity results give a decision table rather than one universal implementation.

Recognized structure	Exact route	Mathematical guarantee
Clause width at most two	MaxSAT / Almost-2-SAT	$\beta = \tau$, sign in P, threshold FPT
Small variable count	Ternary bilateral enumeration	$O(3^k F)$, complete
Bounded incidence treewidth	Tree-decomposition DP via $G(F)$	$2^{O(t)} F ^{O(1)}$
Regular-DIM formula signature	Formula optimization or graph IDS	exact gap $i - \mu^* = \beta$
Unrestricted fixed width $r \geq 3$	proof-logging search, IDS, or MILP	NP-hard already at $q = 0$

This table supports algorithm selection in the precise sense that recognized structure determines applicable exact reductions and parameterized bounds. It does not claim that a prototype router can predict wall-clock dominance among well-engineered solvers.

5.8 Paired-edge normal form for the unresolved $(3, 2, 2)$ slice

The occurrence restriction admits a more precise reformulation than generic 3-CNF. Let F be proper, simple, and exact signed-occurrence $(3, 2, 2)$, with m clauses and k variables. Form a clause multigraph $Q(F)$ whose vertices are the clauses. For each signed literal ℓ , join the two clauses containing ℓ by an edge e_ℓ . The graph is cubic, parallel edges are possible, and for every variable x the two edges e_x^+, e_x^- are vertex-disjoint and form a prescribed pair. Conversely, every cubic multigraph whose edges are partitioned into such disjoint pairs recovers an exact signed-occurrence $(3, 2, 2)$ formula up to variable names and sign switches.

Call a set $S \subseteq E(Q(F))$ **pair-feasible** when it contains at most one edge from each prescribed pair and, for every pair not represented in S , each of its two edges has at least one endpoint outside $V(S)$. Then

$$\beta(F) = \frac{m}{4} - \max_{S \text{ pair-feasible}} (|V(S)| - |S|). \quad (5.1)$$

To prove the identity, map an assigned variable to the edge belonging to its satisfying sign and map an unassigned variable to no edge. The clauses satisfied by assigned literals are exactly $V(S)$. The condition imposed on an unrepresented pair says exactly that at least one occurrence of each sign survives, which is bilaterality. Since $|S| = k - |U|$,

$$|T| - |U| = m - |V(S)| - (k - |S|) = \frac{m}{4} + |S| - |V(S)|,$$

where $3m = 4k$ was used in the last equality. The construction is reversible, so minimization gives (5.1).

Thus the exact sign question is equivalent to deciding whether the paired coverage surplus in (5.1) is at least $m/4$. This resembles a rainbow matching problem because at most one edge may be selected from each two-edge color class, but it has an additional induced-coverage condition on every unselected pair and the objective is $|V(S)| - |S|$, not cardinality. Ordinary rainbow-matching hardness therefore does not, without a further reduction, settle this slice. Nor can the maximum in (5.1) be restricted to pair-feasible matchings: the supplied 9-variable, 12-clause counterexample has $\beta = -1$, hence optimum surplus four, while exhaustive matching search finds no pair-feasible matching larger than three. Overlapping selected edges are therefore an essential part of the exact normal form. The enforcer family of Theorem 12 is a polynomial subcase: its normal form collapses to minimum edge cover. Equation (5.1) records the remaining complexity obstruction exactly and prevents the broader width-three hardness theorem from being misreported as a solution of the bounded-occurrence case.

6. Regular graphs with a dominating induced matching

Let G be a finite simple d -regular graph, $d \geq 2$, supplied with a dominating induced matching M of size k . A dominating induced matching is an induced matching that edge-dominates the graph. Orient each edge of M , label its endpoints as the positive and negative literal vertices of one variable, and turn every vertex outside $V(M)$ into an indexed clause containing its neighboring literal vertices.

This construction has rigid consequences. Endpoints of different matching edges have no edges between them because M is induced. Vertices outside $V(M)$ form an independent set: an edge between two such vertices would share no endpoint with an edge of M , contrary to edge domination. Each outside vertex has d distinct literal neighbors, so each clause contains d distinct literal vertices. A clause may contain both signs of one variable, which is why tautologies must be allowed in the complete coordinatisation. Each signed literal vertex uses one incident edge in M and has $d - 1$ remaining clause incidences. It therefore occurs exactly $d - 1$ times in the formula.

Conversely, begin with an indexed formula in which every clause contains exactly d distinct literal vertices and every signed literal occurs exactly $d - 1$ times. Its formula graph is d -regular. The complementary-pair edges form an induced matching, every other edge meets exactly one of those edges, and the matching is dominating. The two constructions are inverse up to labels and orientations. Hence this is a surjective coordinatisation of the entire family of d -regular graphs equipped with a specified dominating induced matching, rather than a construction of selected examples.

The phrase “equipped with” is essential. A graph can have more than one dominating induced matching, and orienting a matching edge chooses which endpoint represents the positive literal. Reversing one orientation switches the corresponding variable sign throughout the reconstructed formula. These operations may change the labeled formula while preserving an isomorphic formula graph and the value of β , because Theorem 10 identifies that value with a graph-invariant gap. The coordinatisation is canonical relative to a labeled, oriented matching and invariant, for the purpose of the gap, under the resulting sign switches. It does not assert that an unlabeled graph has a unique formula representation.

Counting degrees gives further invariants. If there are m clause vertices, then the $2k$ literal vertices have $2k(d - 1)$ clause incidences and the clauses have dm such incidences. Therefore

$$m = \frac{2k(d - 1)}{d}, \quad |V(G)| = \frac{2k(2d - 1)}{d}, \quad |E(G)| = k(2d - 1).$$

Let $\mu^*(G)$ denote the minimum size of a maximal matching. The specified pair matching is maximal, so $\mu^*(G) \leq k$. Every maximal matching is edge-dominating. In a d -regular graph, one matching edge can dominate at most $2d - 1$ graph edges: itself and the $2(d - 1)$ other edges incident with its endpoints. Since G has $k(2d - 1)$ edges, every maximal matching has at least k edges. Thus $\mu^*(G) = k$.

This lower bound uses only regularity and maximality. The edges of any maximal matching collectively dominate every graph edge, and one selected edge can cover no more than its closed edge-neighborhood of size $2d - 1$. Equality is witnessed by the specified DIM. The matching parameter is therefore fixed throughout the coordinatised class before independent domination is optimized. That separation makes β a coordinate of the entire gap rather than one term in an equation with two unknown graph optima.

Theorem 10 (regular-DIM gap classification). Let F_M be the indexed formula reconstructed from an oriented specified dominating induced matching of a simple d -regular graph G . Then

$$i(G) - \mu^*(G) = \beta(F_M).$$

Proof. The coordinatisation shows that $G = G(F_M)$. Theorem 2 gives $i(G) = k + \beta(F_M)$, while the preceding edge count proves $\mu^*(G) = k$. Subtraction yields the identity. $\square$

Theorem 11 (constructive regular-DIM replacement bound). Let G be a finite simple d -regular graph, $d \geq 2$, equipped with a dominating induced matching. Then

$$i(G) \leq \mu^*(G) + \left\lfloor \frac{2(d-1)}{d 2^d} \mu^*(G) \right\rfloor.$$

In particular,

$$\frac{i(G)}{\mu^*(G)} \leq 1 + \frac{2(d-1)}{d 2^d}.$$

An independent dominating set satisfying the integral bound can be found in deterministic polynomial time.

Proof. Use the coordinatisation above and write $k = |M| = \mu^*(G)$ and $m = 2k(d-1)/d$ for the number of clause vertices. Choose one endpoint of each edge of M independently and uniformly. A clause vertex is not dominated precisely when none of its literal neighbours is selected. If the associated clause is tautological, two of those neighbours are the endpoints of one matching edge, so this probability is zero. Otherwise its d literal neighbours lie on distinct matching edges and exactly one choice on each edge avoids it. Its non-domination probability is therefore 2^{-d} .

If X counts the undominated clause vertices, linearity of expectation gives

$$EX \leq \frac{m}{2^d} = \frac{2k(d-1)}{d 2^d}.$$

Some choice consequently has $X \leq \lfloor 2k(d-1)/(d 2^d) \rfloor$. Add all of its undominated clause vertices to the k selected literal vertices. Clause vertices are mutually nonadjacent, and an added clause vertex has no selected literal neighbour by definition, so the resulting set is independent. Every literal pair is dominated by its selected endpoint, while every clause vertex is either already dominated or is added. Thus the set is independently dominating and has the asserted size.

For the algorithmic statement, expose the endpoint choices one matching edge at a time and select the branch with no larger conditional expectation. The conditional expectation never increases and the final integer value of X is at most the floor of its initial expectation. All clause contributions can be maintained directly from the incidence representation. $\square$

For cubic graphs this yields the concrete inequalities

$$i(G) \leq \mu^*(G) + \left\lfloor \frac{\mu^*(G)}{6} \right\rfloor \leq \frac{7}{6} \mu^*(G).$$

Since a cubic DIM graph in this coordinatisation has $|V(G)| = 10\mu^*(G)/3$, it also gives

$$\frac{i(G) - \mu^*(G)}{|V(G)|} \leq \frac{1}{20}.$$

More generally, the vertex count preceding Theorem 10 converts Theorem 11 into the all-degree density bound

$$\frac{i(G) - \mu^*(G)}{|V(G)|} \leq \frac{d-1}{2^d(2d-1)}. \quad (6.0)$$

The degree-two endpoint can be settled exactly rather than bounded. Every finite simple 2-regular component is a cycle, and a cycle has a dominating induced matching precisely when its length is divisible by three. On C_{3a} , both the independent domination number and the minimum size of a maximal matching are a . Additivity over components therefore gives

$$i(G) = \mu^*(G), \quad C_2^{\text{DIM}} = 1, \quad \lambda_2^{\text{DIM}} = 0.$$

Thus the complete degree-two case and a constructive upper bound for every $d \geq 2$ are known; positive constructions and sharp constants beyond degree three remain separate questions.

Consequently, the sign of $\beta(F_M)$ gives a complete trichotomy for the gap:

$$\begin{array}{l|l} \beta(F_M) > 0 & i(G) > \mu^*(G), \\ \beta(F_M) = 0 & i(G) = \mu^*(G), \\ \beta(F_M) < 0 & i(G) < \mu^*(G). \end{array}$$

For $d = 3$, non-tautological formulas in the coordinatisation are precisely proper exact signed-occurrence $(3, 2, 2)$ -CNFs. Let P be the explicit 15-variable formula of Appendix A. Proposition 14 proves $\beta(P) = 1$; the original formula was supplied by the motivating release [1]. Let N be Lemma 8's component with $\beta(N) = -1$. Both are proper, simple, and exact signed-occurrence $(3, 2, 2)$. Theorem 4 then proves

$$\{\beta(F) : F \text{ proper simple exact signed-occurrence } (3, 2, 2)\text{-CNF}\} = \mathbb{Z}.$$

Indeed, use z copies of P for an integer $z > 0$, $-z$ copies of N for $z < 0$, and $P \sqcup N$ for zero. The corresponding cubic regular-DIM graphs from this additive proof are generally disconnected. The result is a full value spectrum, not a claim that β determines graph isomorphism or every other invariant.

The following construction supplies the missing connected positive amplification. Its finite ingredient is the $E_{3,2,2}$ enforcer published by Zhang, Peitl, and Szeider [4, Appendix A.1]. In the polarity used here it is the ten-clause formula

$$\begin{aligned} E = & (\neg x_1 \vee \neg x_4 \vee \neg x_8) \wedge (\neg x_2 \vee \neg x_6 \vee \neg x_8) \wedge (\neg x_1 \vee \neg x_6 \vee x_8) \\ & \wedge (\neg x_4 \vee \neg x_7 \vee x_8) \wedge (\neg x_3 \vee \neg x_5 \vee x_7) \wedge (\neg x_2 \vee x_5 \vee x_6) \\ & \wedge (\neg x_3 \vee x_4 \vee \neg x_7) \wedge (x_3 \vee \neg x_5 \vee x_6) \wedge (x_2 \vee x_5 \vee x_7) \\ & \wedge (x_2 \vee x_3 \vee x_4). \end{aligned}$$

The distinguished terminal is x_1 . It occurs negatively twice and not positively; every other literal occurs exactly twice in each sign. Define a local cost by counting surviving clauses of E and subtracting only unassigned internal variables $x_2, \dots, x_8$, leaving the terminal charge until gadgets are closed. Requiring bilaterality only for the internal variables gives exactly the following finite terminal signature; these are all feasible entries:

state of x_1	residual terminal signs	minimum local cost
0	$\emptyset$	0
1	$\emptyset$	1
*	$\emptyset$	1
*	$\{-\}$	1.

(6.1)

The table is a finite lemma with two supplied paths. The Python enumerator checks all $3^8 = 6561$ local partial assignments and records a minimizing witness for every entry. Separately, `lean_terminal_signature.lean` hard-codes the ten displayed clauses, independently defines residual survival, internal bilaterality, terminal masks and local cost, and proves by `native_decide` that the complete computed signature equals exactly the four displayed rows. This path has no DIMACS-parser or Python dependency; its translation boundary is the visible clause transcription. Assigning the terminal zero permits cost zero; assigning it one costs one; and leaving it unassigned can supply its negative residual sign at local cost one before the terminal's single global -1 charge is applied.

Theorem 12 (connected edge-cover amplifier). Let R be a finite simple connected cubic graph with r vertices. For every edge $e \in E(R)$, take a fresh copy E_e of the enforcer, identify its terminal x_1 with a variable y_e , and keep all internal variables disjoint. For every vertex $v \in V(R)$, add the positive clause

$$D_v = \bigvee_{e \ni v} y_e.$$

Call the resulting indexed formula $A(R)$. Then $A(R)$ is proper, simple, exact signed-occurrence $(3, 2, 2)$, and its formula graph is connected. Moreover,

$$\beta(A(R)) = q(R) = r - v(R), \quad (6.2)$$

where $q(R)$ is the minimum edge-cover number and $v(R)$ the maximum matching number. Consequently its cubic DIM graph satisfies

$$|V(G(A(R)))| = 40r, \quad \mu^*(G(A(R))) = 12r, \quad i(G(A(R))) = 12r + q(R). \quad (6.3)$$

Proof. A cubic graph has $3r/2$ edges. Each enforcer copy contributes seven fresh internal variables in addition to its edge terminal and ten clauses; the vertex clauses contribute r more clauses. Thus $A(R)$ has $12r$ variables and $16r$ clauses. Every internal signed literal retains its two occurrences from E . Each terminal occurs negatively twice in its

enforcer and positively in the two vertex clauses at the ends of its edge. The clauses have three distinct variables and no two are equal: enforcer clauses from different copies have disjoint internal variables, and a vertex clause cannot equal an enforcer clause. Hence the formula is proper, simple, and exact signed-occurrence $(3, 2, 2)$.

For the value calculation, let $S \subseteq E(R)$ be the set of terminals assigned one. By (6.1), those enforcers contribute one each, terminals assigned zero contribute zero, and a terminal left unassigned has net contribution zero after its delayed terminal charge. A vertex clause survives exactly when its vertex is not incident with S . Moreover, every unassigned terminal can be replaced by zero without changing which vertex clauses survive or increasing the cost. It follows that

$$\beta(A(R)) = \min_{S \subseteq E(R)} (|S| + u_R(S)), \quad (6.4)$$

where $u_R(S)$ is the number of vertices uncovered by S . Every edge cover has objective equal to its cardinality. Conversely, from any S , add one incident edge for each uncovered vertex; the result is an edge cover of size at most $|S| + u_R(S)$. Hence (6.4) equals $q(R)$. Since R has no isolated vertices, the standard edge-cover identity gives $q(R) = r - v(R)$.

The literal--clause incidence supplied by the vertex clauses contains the subdivision incidence graph of the connected skeleton R . Direct inspection of the displayed enforcer shows that every vertex of each copy's formula graph is joined to its terminal pair. Thus all copies join the same connected core, proving connectivity. Finally, the counts above and Theorem 10 give (6.3). $\square$

Taking $R = C_s \square K_2$, $s \geq 3$, gives $r = 2s$ and a perfect matching of size s , so $q(R) = s$. Therefore there is an explicit infinite family of connected cubic DIM graphs G_s with

$$|V(G_s)| = 80s, \quad \mu^*(G_s) = 24s, \quad i(G_s) = 25s, \quad i(G_s) - \mu^*(G_s) = s. \quad (6.5)$$

In particular, their ratio is $25/24$ and their gap density is $1/80$. The edge-cover reduction also allows a sharper choice of skeleton. O and West proved that every connected cubic graph of order r has matching number at least $(4r - 1)/9$, and characterized an infinite equality family H_1 [13]. A canonical sequence $R_t \in H_1$ has

$$r_t = 16 + 18t, \quad v(R_t) = \frac{4r_t - 1}{9}, \quad q(R_t) = \frac{5r_t + 1}{9}. \quad (6.6)$$

Applying Theorem 12 gives connected cubic DIM graphs of order $40r_t$ and gap $(5r_t + 1)/9$. Their gap density tends to $1/72$, and their ratio tends to $113/108$. The initial skeleton has order 16 and gives the exact tuple $(|V|, \mu^*, i) = (640, 192, 201)$, hence ratio $67/64$. The same O--West lower bound shows that $1/72$ is the sharp asymptotic density attainable inside this edge-cover-amplifier construction, even though it need not be sharp for the full cubic DIM class.

If

$$C_3^{\text{DIM}} = \sup_G \frac{i(G)}{\mu^*(G)} \quad \text{and} \quad \lambda_3^{\text{DIM}} = \limsup_{n \rightarrow \infty} \sup_{G: |V(G)| \geq n} \frac{i(G) - \mu^*(G)}{|V(G)|},$$

where G ranges over connected cubic graphs equipped with a DIM, then the finite base P , Theorem 11, and (6.5) give the certified bounds

$$\frac{16}{15} \leq C_3^{\text{DIM}} \leq \frac{7}{6}, \quad \frac{1}{72} \leq \lambda_3^{\text{DIM}} \leq \frac{1}{20}. \quad (6.7)$$

Corollary 13 (minimum counterexample order in the cubic-DIM class). If a finite simple cubic graph G is equipped with a dominating induced matching and satisfies $i(G) > \mu^*(G)$, then $|V(G)| \geq 50$. Equality is attained by the connected graph generated from P in Appendix A.

Proof. Reconstruct the indexed signed-occurrence formula F_M from the specified DIM. If $i(G) > \mu^*(G)$, Theorem 10 gives $\beta(F_M) > 0$. Every complete assignment is bilateral, so F_M must be unsatisfiable. Delete tautological clauses and collapse repeated indexed clauses. Neither operation changes satisfiability, and the result is an unsatisfiable non-tautological 3-CNF in which each signed literal occurs at most twice. Zhang, Peitl, and Szeider proved that every such $(3, 2, 2)$ -formula has at least 20 distinct clauses [4, Theorem 15]. Hence the original indexed formula has $m \geq 20$ clauses. Its occurrence count is $3m = 4k$, while its formula graph has order $2k + m = 5m/2$, so $|V(G)| \geq 50$. Proposition 14 and a direct incidence traversal show that P has positive value and a connected 50-vertex formula graph. $\square$

7. Six uses and an infrastructure for mathematics

7.1 Systematic discovery

Within the proper exact signed-occurrence $(3, 2, 2)$ space, the formula graph is cubic, its complementary pairs form a dominating induced matching, and

$$G(F) \text{ satisfies } i(G) > \mu^*(G) \iff \beta(F) > 0.$$

This turns counterexample discovery into a necessary-and-sufficient formula search: generate formulas under the signed-occurrence constraints, optimize β , retain positive instances, and translate them canonically. Symmetry breaking can quotient variable renamings, sign switches, and clause permutations without altering the objective. The method is systematic because the coordinatisation is surjective for graphs with a specified DIM. It is not an efficient enumeration theorem, and the sign complexity of the exact $(3, 2, 2)$ slice remains unknown.

7.2 Compact proof production

An upper bound $\beta(F) \leq q$ has a compact certificate: a bilateral partial assignment with deficiency at most q . A lower bound $\beta(F) \geq q$ is different. In unrestricted formulas it is a coNP-type obligation and may require exhaustive branch-and-bound, a SAT or MILP proof log, a graph-side independent-domination certificate, or a structural inequality such as Lemma 8. No uniform polynomial-size lower certificate is claimed.

Composition supplies an important exception. For a block-disjoint formula, Theorem 4 reduces the proof to three checks: certify each distinct component, verify the declared variable and clause partition, and add the values. A benchmark containing thousands of copies need not be searched monolithically. The artifact implements this proof rule and binds each CNF to its reconstructed graph by cryptographic hash. Compactness therefore arises from mathematical structure rather than compression of an unexplained solver verdict.

There are consequently three assurance levels for a reported value. A bilateral assignment alone proves an upper bound. An exhaustive receipt records that a particular program searched its declared domain, but its force depends on the implementation and execution environment. A structural proof or independently checkable proof log establishes the lower bound without trusting the optimizer's summary. The current package uses a written occurrence inequality for the negative gadget, a separately implemented exhaustive solver for the positive base instance, an independently checkable LRAT lower-bound certificate for that base, and additive manifests for large composites. The direct LRAT file is checked both by the pinned native checker and by Lean's verified LRAT checker; the formula-to-encoding equivalence remains a displayed mathematical lemma. A JSON receipt records this chain but is not itself the proof.

7.3 Algorithm selection

Section 5 provides a structural dispatcher. Width-two formulas should be sent to MaxSAT or Almost-2-SAT methods because equality with τ is a theorem. Small formulas can use direct ternary enumeration, and small-incidence-width formulas admit tree-decomposition dynamic programming. A recognized regular-DIM signature permits either formula optimization or graph independent domination, with the output interpreted immediately as a gap. Unrestricted width-three instances require general proof-producing optimization unless further structure is detected. This is a mathematical selection layer, not a learned performance model.

7.4 Algebraic conformance benchmark generation

The algebra gives orthogonal benchmark controls. Width lifting changes the uniform clause width while preserving the target value. Indexed replication changes the penalty scale and exposes a MaxSAT primary optimum plus the u^* secondary optimum. Disjoint union adds prescribed values and convolves all solution multiplicities. The positive and negative exact signed-occurrence $(3, 2, 2)$ components generate every integer target while preserving the regular-DIM signature.

The current artifact includes verified target values 3, -2 , 0, and -4 . Its manifests identify component templates, multiplicities, formula hashes, graph hashes, and the additive proof rule. These are algebraic conformance benchmarks because the expected value follows from a theorem and separately checked base components. They test whether a solver or transformation respects prescribed identities. They are not presented as statistically representative samples of industrial SAT distributions or as an empirical performance suite.

7.5 Classification of a structured graph family

Theorem 10 classifies every finite simple d -regular graph supplied with a dominating induced matching. It identifies the precise formula class, proves the matching parameter $\mu^* = k$, and makes β the complete signed gap coordinate. The word “complete” refers to the trichotomy and numerical value of $i - \mu^*$ throughout this equipped family. It does not mean that two formulas with equal β yield isomorphic graphs, nor that the parameter classifies regular graphs without a specified DIM.

7.6 AI-assisted mathematical infrastructure

Bilateral deficiency supports a typed pipeline suited to assisted discovery:

indexed CNF $\rightarrow$ canonical residual objective $\rightarrow$ exact solver $\rightarrow$ certificate or receipt $\rightarrow$ independent checker $\rightarrow$ graph-theo

The type boundaries matter. Indexed semantics prevent silent loss of duplicate clauses. The bilateral condition is checked independently of the objective. The polynomial identity permits cross-model comparison of every solution size. Composition manifests separate base proofs from large-instance assembly. Tests under both ordinary Python and optimization mode guard against assertion-only verification. During development, a cross-check exposed a specific implementation error: residual polarities had been recorded from a clause before the scan later discovered that another assigned literal satisfied it. The C++ solver was corrected to classify clause survival before recording residual occurrences. This failure is informative because it shows why independently phrased semantics and cross-model tests are necessary.

The positive-base development adds a second instructive failure boundary. A first cardinality encoding was mathematically adequate but generated a large, symmetry-heavy proof. Replacing it by a deterministic prefix-balance automaton produced a compact encoding whose direct LRAT proof is accepted by both a native checker and Lean. An intermediate LRAT translation passed the native checker but not Lean; retaining separate proof paths exposed this format-level incompatibility instead of concealing it behind a solver status. Hashes bind the input, encoding, map, proof, wrapper, and receipt, but hashes establish identity rather than correctness.

The infrastructure claim is technical. The artifact demonstrates that a model can propose instances, route exact methods, generate a lower-bound encoding, and emit checkable evidence without making its own prose authoritative. It does not establish that generated text is a proof, that the complete theorem has been formalized, or that the parameter has received mathematical community adoption.

8. Reproducibility, limitations, and open problems

The accompanying package contains a separately implemented Python semantics, a dependency-free C++ exhaustive solver, unit and exhaustive cross-model tests, a structural analyzer, an algebraic conformance benchmark generator and verifier, a deterministic SAT threshold encoder, three finite threshold encodings, four native-checked LRAT derivations, Lean wrappers for the three direct derivations, and the parser-independent Lean terminal lemma. The Python tests compare bilateral assignments with graph independent dominating sets, including full size spectra, over more than one hundred small indexed formulas. They cover empty, unit, binary, tautological, duplicate-clause, and absent- variable cases. Tests also exercise width-two equality, additivity, convolution, lifting, replication, DIM signatures, graph reconstruction, and mutation rejection. The same suite passes with ordinary execution and `python -O`.

The C++ solver enumerated all $3^{15} = 14,348,907$ partial assignments of the canonical formula, found 939,975 bilateral assignments, and returned $\beta = 1$. It independently returned -1 for N . These receipts verify finite computations and agreement between implementations. They do not prove the universal theorems. Conversely, the written proofs do not certify that a particular source archive or executable was transmitted without corruption. The supplement supplies commands, expected results, and hashes so these assurance layers remain separate. In addition, the lower-bound encoding for P has 731 variables and 9,256 clauses. A native checker verifies both the trimmed and direct LRAT derivations; Lean 4.32.1's `Std.Tactic.BVDecide.LRAT` checker verifies the direct derivation.

Proposition 14 states the encoding lemma and the exact hashes needed to connect these finite artifacts to $\beta(P) = 1$.

The semantic bridge for this first encoding is also checked by a standard- library clean-room validator that imports neither the production encoder nor `bd_core.py`. It independently reconstructs the complete 731-variable map and the multiset and canonical order of all 9,256 clauses. On a seven-formula corpus it compares the mathematical ternary-state oracle with an independent DPLL implementation at the exact thresholds $q = \beta - 1$ and $q = \beta$, including 66

fixed-state checks, and rejects a rehashed clause mutation. The production compiler is invoked only as a black box to make the corpus encodings. This is producer-side clean-room conformance evidence, not an unaffiliated reproduction or a proof of the universal encoding lemma.

The enforcer composition used to validate the terminal calculus has a separate certificate chain. Its independently regenerated threshold encoding again has 731 variables and 9,256 clauses. The direct LRAT proof has 35,737 actions and is accepted by both the pinned native checker and Lean. This second proof is not needed for Corollary 13 or the integer spectrum, which already use P ; it checks an attributed, structurally distinct positive base and the generic threshold compiler on a second input.

The connected occurrence switch is an extended, optional third finite certificate chain; no theorem in this paper depends on replaying it. Its value-one witness is checked directly, while a byte-regenerated 2,686-variable, 65,061-clause encoding of $\beta \leq 0$ is refuted by a 5,793,599,477-byte direct LRAT proof. The pinned native checker reports VERIFIED; Lean parses 11,393,023 actions and reports LEAN_LRAT_VERIFIED. The proof's size is an engineering limitation, not a logical gap, and motivates compact structured certificates for future connected searches. The core archive retains its formula, encoding, map, hash and verification receipt; the raw 5.79 GB LRAT is distributed as a separately listed extended object.

Three mathematical problems are immediate. The complexity of $\beta(F) \leq 0$ on proper simple exact signed-occurrence $(3, 2, 2)$ -CNF is open. Theorem 12 settles connected positive amplification, but the exact constants in (6.7), and connected constructions for arbitrary signed values remain open. Corollary 13 settles the minimum order within the cubic-DIM class, although classification of all order-50 extremizers and the minimum outside DIM remain open. The general threshold compiler supplied with the artifact should be extended with compact proof-producing backends for larger instances. Formalization of the encoding lemma and Theorems 1, 2, 7, 10, 11, and 12 is also desirable.

Two external limitations remain. The priority search was bounded and should be extended through MathSciNet, zbMATH, dissertations, and alternative terminology. No empirical solver competition has been conducted, so the algorithm table records applicability and asymptotic guarantees rather than performance rankings. The work is unrefereed. Three finite threshold encodings have direct LRAT derivations checked by Lean, and the enforcer terminal table has a separate parser-independent Lean proof, but the general mathematical theory has not been formalized in a proof assistant.

These limits also constrain the novelty claim. The search found a close antecedent that materially changed the paper's wording, which shows that the audit was substantive. It does not show that every terminology branch has been exhausted. Mathematical correctness, computational reproduction, historical priority, and scholarly acceptance are four distinct review questions and are reported separately here.

9. Conclusion

Bilateral deficiency satisfies the criteria set in the introduction. It is the minimum deficiency of a bipolar residual, the exact formula-side coordinate of independent domination on formula graphs, an additive and width-stable optimization parameter, equal to the width-two MaxSAT defect and capable of recovering general MaxSAT after replication. Its sign changes from polynomial-time decidable at width two to NP-complete at every fixed uniform clause width at least three when occurrences are unrestricted. On regular graphs with a specified dominating induced matching, it is exactly the invariant $\text{gap } i - \mu^*$, and its proper exact signed-occurrence $(3, 2, 2)$ spectrum is all integers. The enforcer construction additionally gives an infinite connected cubic family with linear gap and identifies its value exactly with an edge-cover number. O--West skeletons yield asymptotic gap density $1/72$, while the SAT 2024 clause lower bound proves that order 50 is minimum among cubic graphs admitting a dominating induced matching and satisfying $i(G) > \mu^*(G)$. The minimum outside the DIM-equipped class remains open.

The strongest warranted originality claim is correspondingly precise. The partial-assignment cost decomposition has a 2008 antecedent [2]. The explicit bilateral domain, residual parameter, size-preserving bijection, algebra, complexity boundary, regular-DIM coordinatisation, and reproducible infrastructure constitute the advance supported here. These results establish that bilateral deficiency is not disposable notation for one graph. Global priority, full formal verification, peer acceptance, and adoption remain separate questions.

The resulting research programme is concrete. Complexity on the exact signed-occurrence $(3, 2, 2)$ sign slice, sharp regular-DIM constants, classification of the order-50 extremizers, stronger proof-producing compilation, and proof-assistant formalization of the theory are independent next steps. None is needed for the parameter identity, but each would strengthen a different assurance layer. In its current form, the theory already supplies a general definition, exact translations, separations, closure operations, a sharp width boundary, and a complete regular-DIM gap coordinate. That combination is the mathematical content of the claim that β is a parameter in its own right.

Appendix A. The positive exact signed-occurrence (3, 2, 2) base

This appendix makes the only finite positive component used in the integer spectrum theorem completely explicit. Let P be the following indexed CNF; the displayed order is its clause indexing.

$$\begin{aligned}
 & (x_1 \vee x_9 \vee \neg x_{11}) \wedge (\neg x_4 \vee \neg x_{10} \vee \neg x_{13}) \wedge (x_1 \vee \neg x_9 \vee \neg x_{14}) \\
 & \wedge (x_2 \vee x_6 \vee x_{14}) \wedge (\neg x_5 \vee \neg x_6 \vee x_{15}) \wedge (x_2 \vee \neg x_3 \vee x_{15}) \\
 & \wedge (x_3 \vee x_5 \vee \neg x_9) \wedge (\neg x_6 \vee \neg x_{12} \vee \neg x_{15}) \wedge (\neg x_1 \vee x_5 \vee \neg x_{11}) \\
 & \wedge (x_4 \vee \neg x_7 \vee \neg x_{15}) \wedge (\neg x_2 \vee x_8 \vee \neg x_{12}) \wedge (x_3 \vee x_9 \vee x_{11}) \\
 & \wedge (\neg x_4 \vee \neg x_8 \vee x_{10}) \wedge (x_7 \vee \neg x_8 \vee x_{13}) \wedge (x_4 \vee x_7 \vee \neg x_{13}) \\
 & \wedge (\neg x_2 \vee \neg x_7 \vee x_{14}) \wedge (\neg x_3 \vee x_{11} \vee \neg x_{14}) \wedge (x_8 \vee x_{10} \vee x_{12}) \\
 & \wedge (\neg x_{10} \vee x_{12} \vee x_{13}) \wedge (\neg x_1 \vee \neg x_5 \vee x_6).
 \end{aligned}$$

Every clause has three distinct variables and the clauses are pairwise different. The complete signed-occurrence audit is:

variable	positive clause indices	negative clause indices
x_1	1, 3	9, 20
x_2	4, 6	11, 16
x_3	7, 12	6, 17
x_4	10, 15	2, 13
x_5	7, 9	5, 20
x_6	4, 20	5, 8
x_7	14, 15	10, 16
x_8	11, 18	13, 14
x_9	1, 12	3, 7
x_{10}	13, 18	2, 19
x_{11}	12, 17	1, 9
x_{12}	18, 19	8, 11
x_{13}	14, 19	2, 15
x_{14}	4, 16	3, 17
x_{15}	5, 6	8, 10

Thus P is proper, simple, and exact signed-occurrence (3, 2, 2).

Proposition 14 (checker-assisted finite base). $\beta(P) = 1$.

Proof. For the upper bound, use the ternary assignment

$$\alpha = 0000 \text{ * * * * *},$$

where the first four variables are assigned zero and the other eleven are unassigned. Exactly the indexed clauses

$$1, 3, 4, 5, 7, 8, 10, 12, 14, 15, 18, 19$$

survive. Each unassigned variable occurs in both signs among those twelve clauses, so α is bilateral and $\delta_P(\alpha) = 12 - 11 = 1$. Hence $\beta(P) \leq 1$.

For direct inspection, the positive/negative residual witnesses for $x_5, \dots, x_{15}$ are respectively

variable	positive clauses	negative clauses
x_5	7	5
x_6	4	5, 8
x_7	14, 15	10
x_8	18	14
x_9	1, 12	3, 7
x_{10}	18	19
x_{11}	12	1
x_{12}	18, 19	8
x_{13}	14, 19	15
x_{14}	4	3
x_{15}	5	8, 10

For the lower bound, encode the proposition $\beta(P) \leq 0$ as CNF. For each x_j , use three Boolean variables z_j^0, z_j^1, u_j , constrained to have exactly one true value, representing assignment to zero, assignment to one, and nonassignment. For every indexed clause C_a , define a Boolean variable r_a by clauses expressing

$$r_a \leftrightarrow \bigwedge_{\ell \in C_a} \neg s_\ell, \quad s_{x_j} = z_j^1, \quad s_{\neg x_j} = z_j^0.$$

Thus r_a is true exactly when C_a survives. The implications

$$u_j \rightarrow \bigvee_{a: x_j \in C_a} r_a, \quad u_j \rightarrow \bigvee_{a: \neg x_j \in C_a} r_a$$

express bilaterality. Finally, order the bits as $r_1, \dots, r_{20}, u_1, \dots, u_{15}$, with respective weights $+1$ and -1 . One-hot variables $q_{i,d}$ track the unique prefix sum. Starting from $q_{0,0}$, the clauses encode both transitions

$$q_{i-1,d} \wedge \neg b_i \rightarrow q_{i,d}, \quad q_{i-1,d} \wedge b_i \rightarrow q_{i,d+w_i},$$

and forbid every final state with $d > 0$.

We record the required encoding lemma explicitly. The encoding is satisfiable if and only if there is a bilateral partial assignment γ with $\delta_p(\gamma) \leq 0$. In the forward direction, the exactly-one clauses define γ ; the r_a equivalences identify its indexed residual, the two polarity implications give bilaterality, and induction over the automaton layers shows that the final state equals $\sum_a r_a - \sum_j u_j \leq 0$. Conversely, any such γ fixes the assignment and residual indicators, and setting the unique true state in each layer to the actual prefix sum satisfies every encoding clause. This proves the equivalence.

The resulting DIMACS instance has 731 variables and 9,256 clauses. Its SHA-256 digest is c282d2d35f66e81ae175d9afa560b9a5ded2d00f51d8b02c380d939102156c11. The trimmed companion certificate is obtained through the DRAT proof format [14]. The independently checkable lower-bound derivations use LRAT [15] and were generated by CaDiCaL 3.0 [16]. The direct LRAT derivation has digest 61bb30cadaf1cbf4bd15662a1a97b0a6352b416a187532c887000493f46f755a. The pinned native `lrat-check` accepts it, and Lean 4.32.1 [17] using the LRAT checker now distributed in Lean core [18] accepts all 118,497 actions and returns `LEAN_LRAT_VERIFIED`. Therefore the encoded CNF is unsatisfiable. By the encoding lemma, no bilateral assignment has value at most zero, so $\beta(P) \geq 1$. Together with the displayed witness, $\beta(P) = 1$. $\square$

The assurance boundary is exact. The LRAT checker proves unsatisfiability of the 9,256-clause encoding, and Lean's `LRAT.check_sound` theorem connects a successful LRAT check to `CNF.Unsat`. A clean-room validator independently reconstructs the symbolic map and every clause of this encoding and performs small-formula semantic checks, reducing - but not eliminating - the translation obligation represented by the written encoding lemma. The general bilateral-deficiency theory is not thereby fully formalized. The artifact regenerates the CNF and encoding map byte-for-byte, checks the witness independently, invokes both native LRAT paths and the Lean path, and emits deterministic receipts.

Declarations

Data and code availability

The manuscript is accompanied by a reproducibility supplement and an artifact archive containing source code, indexed CNF instances, generated graph data, tests, and exact receipts. Version 1.0.1-candidate is archived at <https://doi.org/10.5281/zenodo.21857209>, with source at <https://github.com/ipitchford/bilateral-deficiency>. The motivating TxGraffiti release and its public repository are listed in Reference [1].

Ethics approval

This theoretical and computational study involved no human participants, personal data, animals, or biological materials. Institutional ethics approval was therefore not applicable.

Author contributions (CRediT)

Anonymous: Conceptualization, Formal analysis, Methodology, Software, Validation, Writing-original draft, and Writing-review and editing. Ian Pitchford acts as archive maintainer and Evidence Press publisher and is not listed as a scholarly author. AI systems are not authors.

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